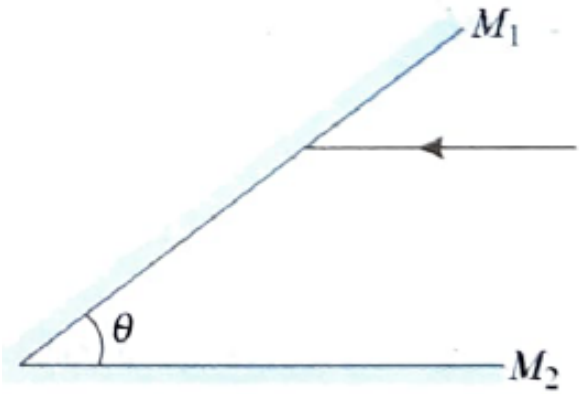
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two plane mirrors M_1 and M_2 are inclined at angle θ. A light ray incident travelling parallel to the surface of mirror M_2 and incident on mirror M_1. What should be the value of θ so that light ray retraces its path after third reflection.



(A) 30 $^\circ$

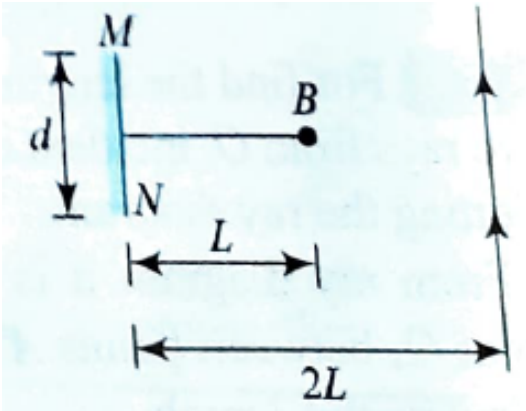
(B) 60 $^\circ$

(C) 36 $^\circ$

(D) 40 $^\circ$

Q.2

A point source of light B is placed at a distance L in front of the centre of a mirror of width d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it, as shown in figure. Find the greatest distance over which he can see the image of the light source from the mirror.



(A) 1d

(B) 5d

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(C) 3d

(D) 4d

Q.3

A driver stops his car at a red light. The car is fitted with side view mirror of focal length $10\ \mathrm{m}$. An ambulance is approaching the car at a constant speed of $16\ \mathrm{ms}^{-1}$. The speed of image of the ambulance, as seen by the driver in the side view mirror, when it is at distance of $50\ \mathrm{m}$ from the mirror will be

(A) $1\ \mathrm{ms}^{-1}$

(B) $\frac{4}{9}\ \mathrm{ms}^{-1}$

(C) $\frac{2}{3}\ \mathrm{ms}^{-1}$

(D) $2\ \mathrm{ms}^{-1}$

Q.4

A person walks at a velocity v in a straight line forming an angle α with the plane of a plane mirror. Determine the velocity v_{rel} at which he approaches his image, assuming that the object and its image are symmetric relative to the plane of the mirror.

(A) $2v\sin\alpha$

(B) $2v\cos\alpha$

(C) $v\sin\alpha$

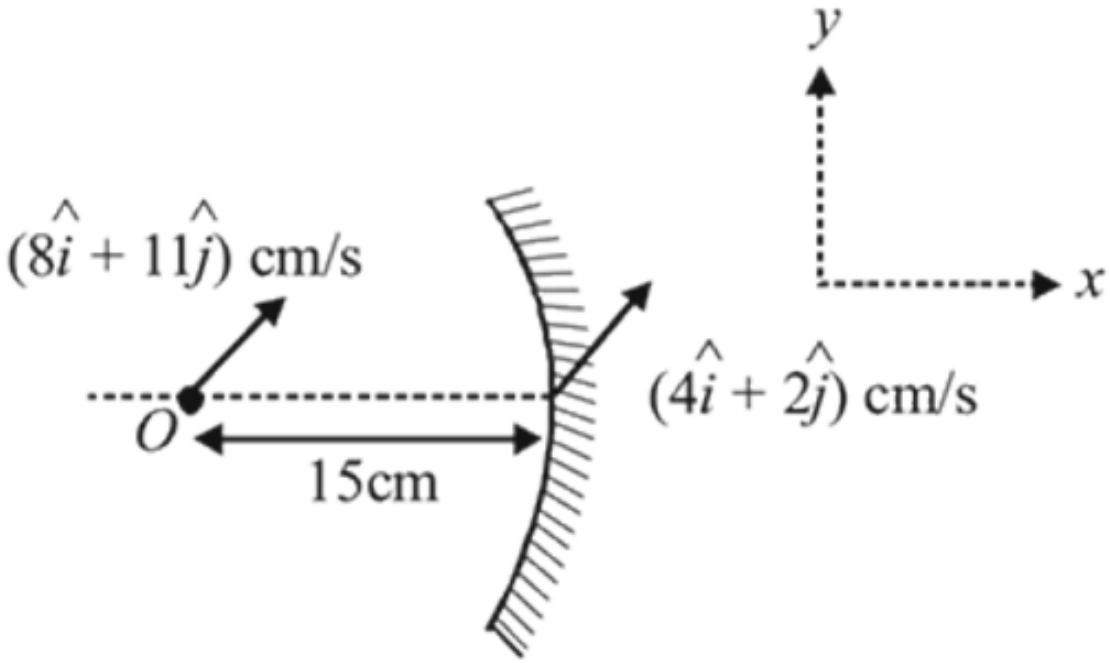
(D) $v\cos\alpha$

Q.5

A point object is located at a distance $15\ \mathrm{cm}$. from the pole of a concave mirror of focal length $10\ \mathrm{cm}$ on its principal axis is moving with a velocity $(8\hat{i}+11\hat{j})\mathrm{cm/s}$ and velocity of mirror is $(4\hat{i}+$

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$2\hat{j})\mathrm{cm/s}$ as shown. If $\vec{v}$ is the velocity of image. Then find the value of $|\vec{v}|$ in $(\mathrm{cm/s})$.



(A) 20

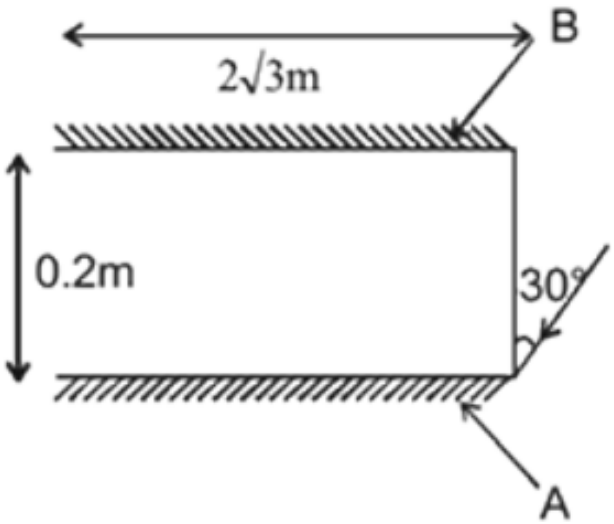
(B) 30

(C) 10

(D) 40

Q.6

Two plane mirrors A and B are aligned parallel to each other, as shown in the figure. A light ray is incident at an angle 30° at a point just inside one end of A. The plane of incidence coincides with the plane of the figure. The maximum number of times the ray undergoes reflections (including the first one) before it emerges out is



(A) 28

(B) 30

(C) 32

(D) 34

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Q.7

When an object is placed at a distance of $25\ \text{cm}$ from a mirror, the magnification is m_1. The object is moved $15\ \text{cm}$ further away with respect to the earlier position, and the magnification becomes m_2. If $m_1/m_2 = 4$, the focal length of the mirror is:

(A) $10\ \text{cm}$

(B) $30\ \text{cm}$

(C) $15\ \text{cm}$

(D) $20\ \text{cm}$

Q.8

The width of man's face is $10\ \text{cm}$. The distance between the eyes of the man is $4\ \text{cm}$. Then the minimum width of plane mirror to see his full face, is

(A) $5\ \text{cm}$

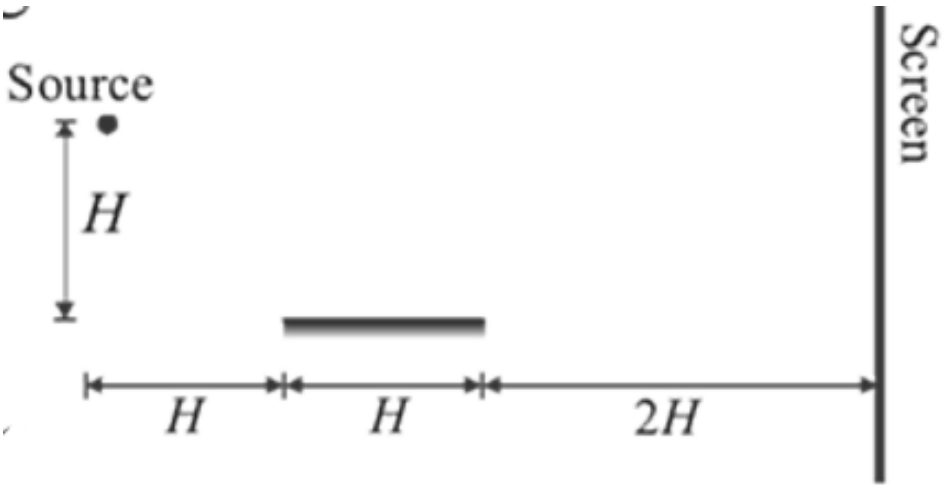
(B) $4\ \text{cm}$

(C) $3\ \text{cm}$

(D) $10\ \text{cm}$

Q.9

A point source has been placed as shown in the figure. What is the length on the screen that will receive reflected light from the mirror?



(A) $2H$ Source

(B) $3H$

(C) H

(D) None of these

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Q.10

A point object is kept in front of a plane mirror. The plane mirror is doing SHM of amplitude 2 cm. The plane mirror moves along the $\mathbf{x}$ - axis which is normal to the mirror. The amplitude of the mirror is such that the object is always in front of the mirror. The amplitude of SHM of the image is

(A) 0

(B) $2\ \mathrm{cm}$

(C) $4\ \mathrm{cm}$

(D) $1\ \mathrm{cm}$

Q.11

A point object is moving with velocity $\vec{u} = 2\hat{i} + \hat{j} - \hat{k}\ \mathrm{m/s}$ in front of a stationary plane mirror. The magnitude of relative velocity of the image with respect to object is maximum if the normal of the plane mirror will be along

(A) $2\hat{i} + \hat{j} + \hat{\hat{k}}$

(B) $-2\hat{i} + \hat{j} - \hat{k}$

(C) $2\hat{i} + \hat{j} - \hat{k}$

(D) $2\hat{i} - \hat{j} - \hat{k}$

Q.12

A ray of light is incident at 50° on the middle of one of the two plane mirrors arranged at an angle of 60° between them. The ray then touches the second mirror, get reflected back to the first mirror, making an angle of incidence of

(A) 50°

(B) 60°

(C) 70°

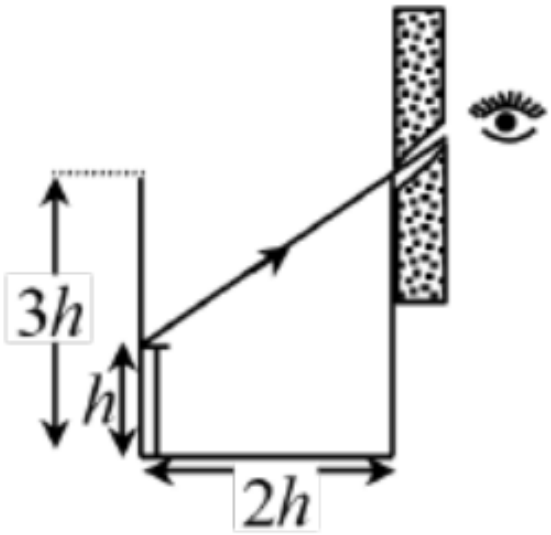
(D) 80°

Q.13

An observer can see through a pinhole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is $3h$ and its radius h. When the beaker is filled with a liquid upto a height $2h$, he can see the lower

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end of the rod. Then the refractive index of the liquid is



(A) $5/2$

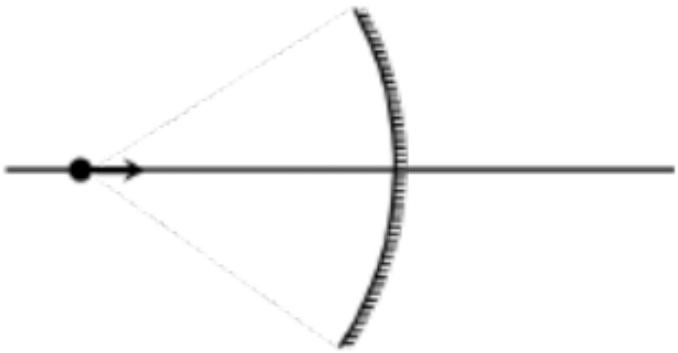
(B) $\sqrt{5/2}$

(C) $\sqrt{3/2}$

(D) $3/2$

Q.14

A point object is moving along the principle axis of a concave mirror at rest of focal length $30\ \text{cm}$ with speed $5\ \text{m/s}$ towards the mirror. Find the speed of image of object when object is at a distance $60\ \text{cm}$ from minor.



(A) $5\ \text{m/s}$ away from minor

(B) $5\ \text{m/s}$ towards the mimor

(C) $10\ \text{m/s}$ away from minor

(D) $10\ \text{m/s}$ towards the mirror

Q.15

Let the focal length of the objective and eye lens be 1.2 cm and 3 cm, respectively. Let the object is

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put 1.25 cm away from the objective lens and the final image is formed at infinity. What is the magnifying power of the microscope?

(A) 150

(B) 200

(C) 250

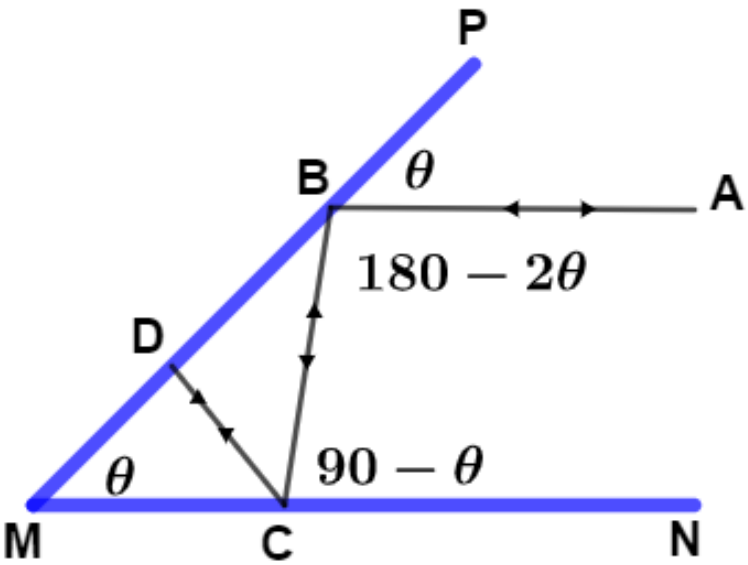
(D) 400

Answers & Solutions

Q.1 Answer:

30°

Solution:



From the diagram's geometry, we see that,

$\angle CDM = 90^\circ$

So in right angle triangle CDM we have,

$\angle DCM = 90 - \theta$ **which is the angle of reflection at point** C

and

$\angle BCN = \angle DCM = 90 - \theta \rightarrow$ **(i) angle of incidence at point** C **equals to the angle of reflection.**

Also,

$\angle PBA = \theta$ **which is the angle of incidence at point** B

$\angle CBD = \angle PBA = \theta$ **angle of reflection equals to angle of incidence at point** B.

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Now for straight-line PM,

$\angle PBA + \angle ABC + \angle CBD = 180^\circ$

On putting the values we get,

$\angle ABC = 180 - 2\theta \rightarrow$ **(ii)**

Now, we see that lines

$\mathrm{AB}||\mathrm{MN}$ **and** BC **is an intercept. So, by the property of the geometry of lines we know that the sum of angles on the same side by the intercept line between two lines is** 180° **which can be written as**$\angle \mathrm{ABC} + \angle \mathrm{BCN} = 180^\circ$

On putting the value of these angles from equations (i) and (ii), we get

$180 - 2\theta + 90 - \theta = 180$

$\Rightarrow 3\theta = 90$

$\therefore \theta = 30^\circ$

Q.2 Answer:

3d

Solution:

Draw the rays BM **and** BN **incident on the mirror edges, and draw the reflected rays (see figure). From the figure, it is clear that the man can observe in the angular region** $P'B'Q'$**. The observer can see through the distance** $P'Q'$**, where reflected ray from** B **can meet the line.**

From geometry, $CD = d = C'D'$

$BD = DP = d/2; CB = QC = d/2$

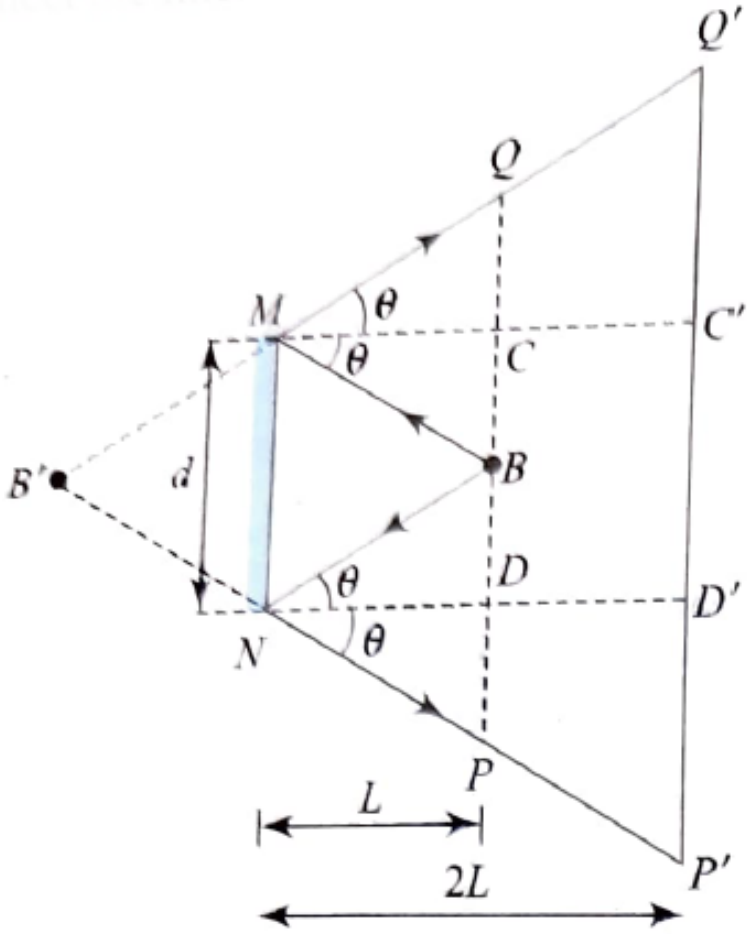
$P'Q' = CD' + P'D' + Q'C'$

$\Rightarrow \quad \frac{ND}{DP} = \frac{ND'}{D'P'} \Rightarrow \frac{L}{d/2} = \frac{2L}{D'P'} \Rightarrow D'P' = d$

Similarly, $Q'C' = d$ **Hence, distance through which the man can observe the image is**

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$Q'C' + C'D' + D'P' = d + d + d = 3d$



Q.3 Answer:

$\frac{4}{9}\ \text{ms}^{-1}$

Solution:

The side view mirror of a car is a convex mirror. Given $f = +10$ m and $u = -50$ cm. Using these values in $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we have

$\frac{1}{v} + \frac{1}{-50} = +\frac{1}{10} \Rightarrow v = +\frac{50}{6}$ m

Differentiating $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ with respect to t, we have

$-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$

$\Rightarrow \quad \frac{dv}{dt} = -\left(\frac{v^2}{u^2}\right)\frac{du}{dt}$

or $\quad \left|\frac{dv}{dt}\right| = \frac{v^2}{u^2} \times \frac{du}{dt}$

$\therefore$ Speed of image $= \frac{v^2}{u^2} \times$ speed of ambulance (object)

$= \frac{(50/6)^2}{(50)^2} \times 16$

$= \frac{4}{9}\ \text{ms}^{-1}$

Q.4 Answer:

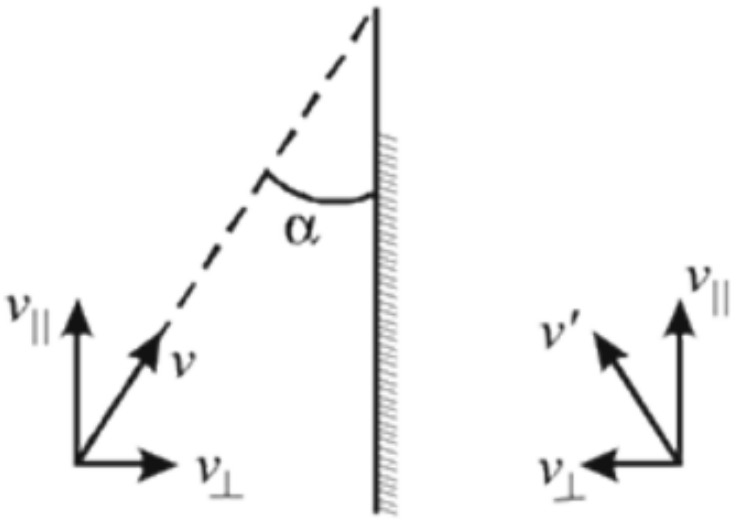
$2v \sin\alpha$

Solution:

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We resolve the velocity vector $\vec{v}$ of the person into two components, one parallel to the mirror, $\vec{v}_{\|}$ and the other perpendicular to the mirror, $\vec{v}_{\perp}$, i.e. $\vec{v}=\vec{v}_{\|}+\vec{v}_{\perp}$(figure). The velocity of the image will obviously be $\vec{v}'=\vec{v}_{\|}-\vec{v}_{\perp}$. Therefore

the velocity at which the person approaches his image is defined as his velocity relative to the image. From the formula $v_{\text{rel}}=2v_{\perp}=2v\sin\alpha$.



Q.5 Answer:

20

Solution:

$\vec{v}_{\text{obj, mirror}}=4\hat{i}+9\hat{j}\quad \frac{dx}{dt}=4,\frac{dy}{dt}=9; u=-x$

$-\frac{1}{10}=\frac{1}{V}+\frac{1}{-x}; V=\frac{-10x}{x-10}$

$v_{1x}=\frac{dV}{dt}=\left(\frac{10}{x-10}\right)^2\left(\frac{dx}{dt}\right); v_{1x}=-16$

$m=-\frac{V}{-x}=\frac{-10}{x-10}=\frac{y_1^0}{y}$

$v_{1y}=-\left(\frac{10}{x-10}\right)\frac{dy}{dt}; v_{1y}=-18$

$\vec{v}_{\text{image , mirror}}=-16\hat{i}-18\hat{j};\quad \vec{v}_{\text{image}}=-12\hat{i}-16\hat{j}$

$v\mid=20\ \text{cm/s}$

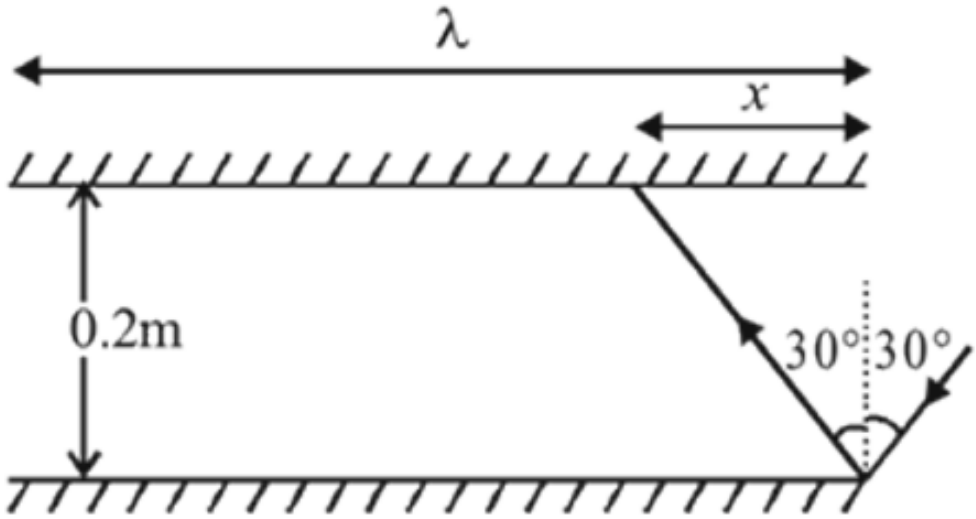
Q.6 Answer:

30

Solution:

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Maximum number of reflection $= \frac{2\sqrt{3}}{x}$ **where** $x = 0.2 \tan 30^\circ = 0.2/\sqrt{3}$.



Q.7 Answer:

20 cm

Solution:

$$m = -\frac{v}{u} = -\left(\frac{f}{u-f}\right)$$

$$\text{Now } m_1 = -\left(\frac{f}{25-f}\right)$$

$$\text{and } m_2 = -\left(\frac{f}{40-f}\right)$$

$$\therefore \quad \frac{m_1}{m_2} = \frac{40-f}{25-f} \text{ or } 4 = \frac{40-f}{25-f} \text{ or } f = 20 \text{ cm}.$$

Q.8 Answer:

3 cm

Solution:

Min. width of plane mirror to see full face $= \frac{D-d}{2} = \frac{10-4}{2} = 3$ cm

Q.9 Answer:

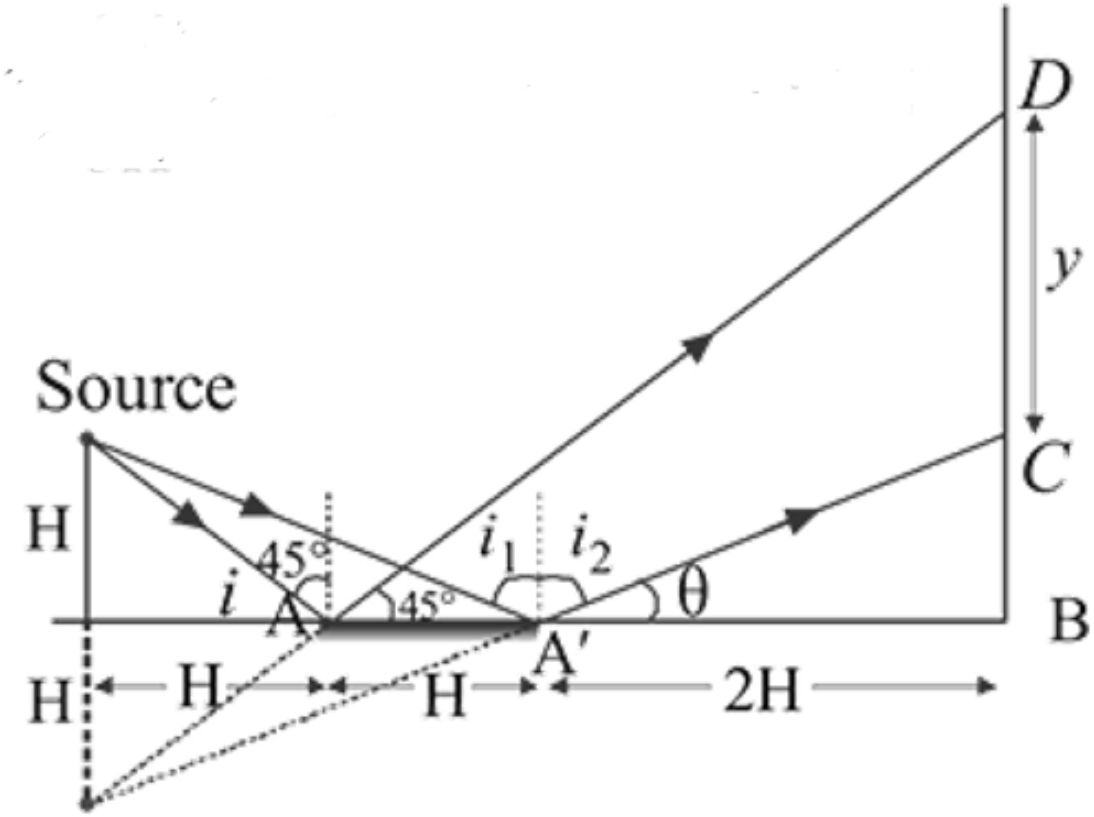
$2H$ Source

Solution:

$$\text{In } \Delta ABD, \frac{BD}{3H} = \tan 45^\circ \text{ or } BD = 3H$$

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In $\Delta ABD, \frac{BD}{3H} = \tan 45^\circ$ or $BD = 3H$

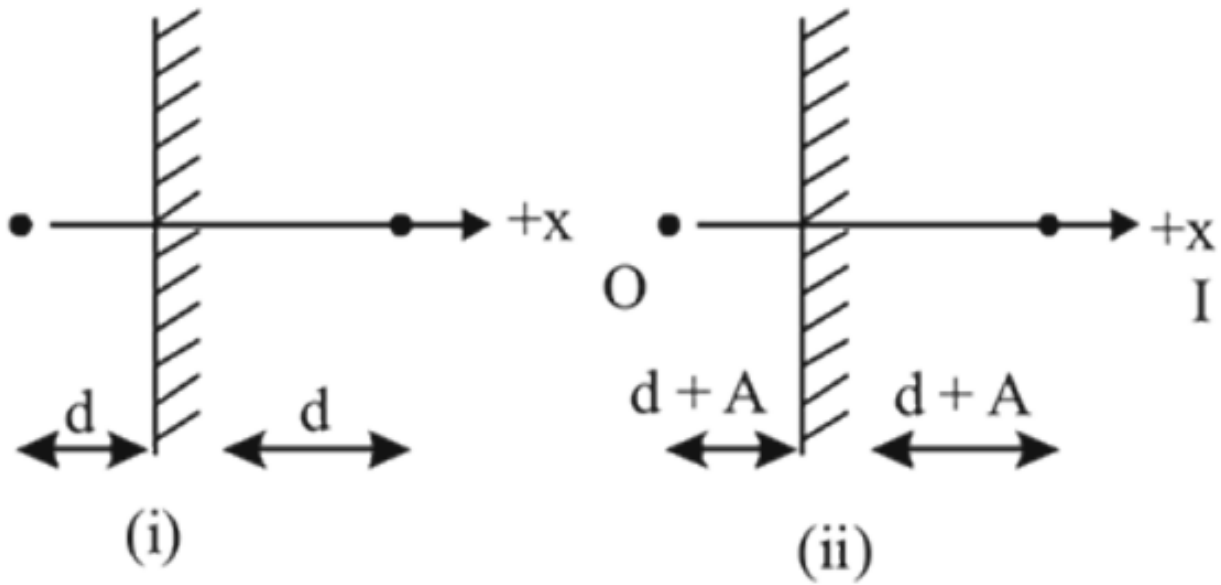


And in $\Delta A'BC, \frac{BC}{2H} = \tan\theta = \frac{1}{2}$ or $BC = H$
Now, $y = BD - BC = 3H - H = 2H$.

Q.10 Answer:

4 cm

Solution:



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From Fig. (i) and (ii), its is clear that if the mirror moves distance 'A', then the image moves a distance $'2\ \mathrm{A}'$

Q.11 Answer:

$2\hat{i}+\hat{j}-\hat{k}$

Solution:

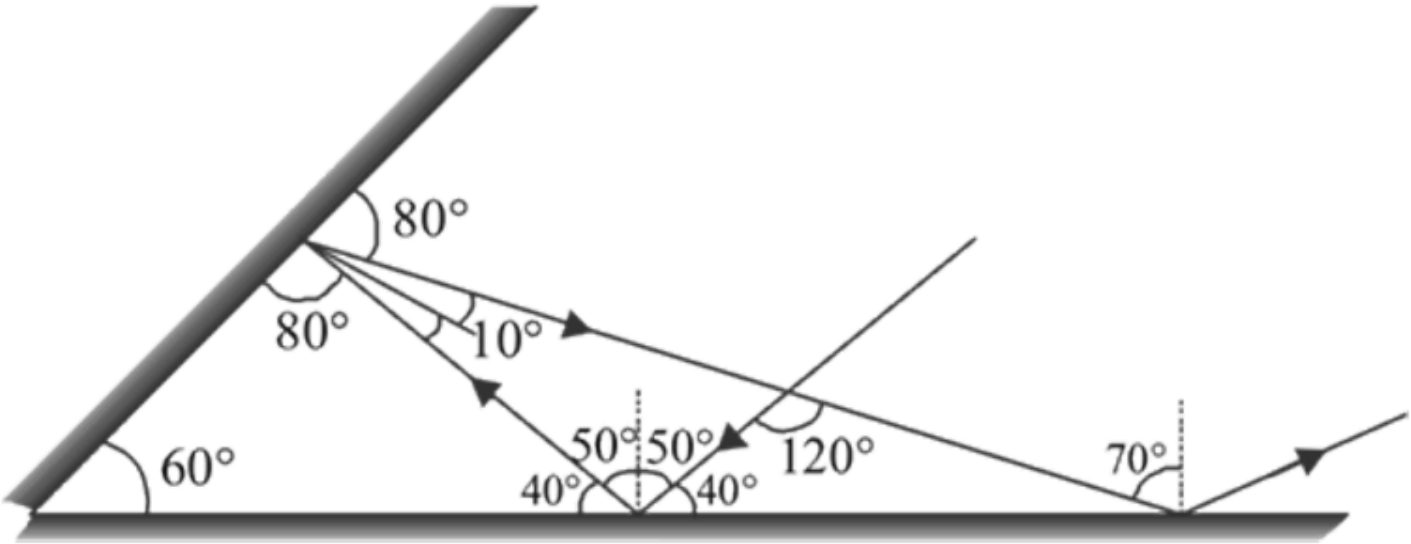
No solution available

Q.12 Answer:

70°

Solution:

See geometry of the figure.



Q.13 Answer:

$\sqrt{5/2}$

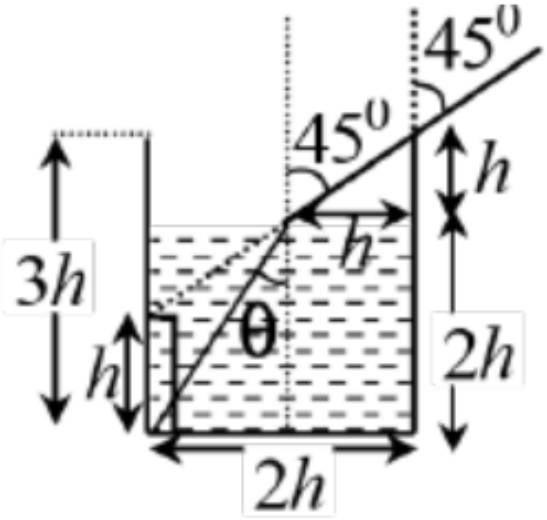
Solution:

$\mu \sin\theta = \sin 45^\circ$

$\mu \frac{h}{\sqrt{5}} h = \frac{1}{\sqrt{2}}$

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$\mu = \sqrt{\frac{5}{2}}$



Q.14 Answer:

$5\ \mathrm{m/s}$ away from minor

Solution:

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, u = -60 f = -30$

$\frac{1}{-60} + \frac{1}{v} = -\frac{1}{30}, \frac{1}{v} = -\frac{1}{30} + \frac{1}{60} = -\frac{1}{60}$

$v = -60\ \mathrm{cm}$ **and** $\frac{1}{v^2}\frac{dv}{dt} + \frac{1}{u^2}\frac{du}{dt} = 0, \frac{du}{dt} = -\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$

$\frac{dv}{dt} = -5\ \mathrm{m/s} \quad \therefore$

Q.15 Answer:

200

Solution:

The microscope here, is a compound microscope which is a combination of objective and eyepiece.

For the objective lens,

$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

Objective lens is convex, so $f_o = 1.2\ \mathrm{cm}$

$\frac{1}{+1.2} = \frac{1}{v_o} - \frac{1}{(-1.25)} \frac{1}{+1.2} = \frac{1}{v_o} + \frac{1}{1.25}\frac{1}{v_o} = \frac{1}{1.2} - \frac{1}{1.25} v_o = 30\ \mathrm{cm}$

When final image is at infinity,

Net magnification, $m = m_o \times m_e$

$m = \frac{v_o}{u_o} \times \frac{D}{f_e}$

$m = \frac{30}{-1.25} \times \frac{25}{3} m = 200$

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